

Replication Report: Jouvét (2023) “Inversion of a Stokes glacier flow model emulated by deep learning”

IGM v3.1.1 data-assimilation on Grosser Aletsch Gletscher
TIER-1 GAP-FILL for AI ATLAS P021 (ice-sheet basal friction inversion)

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1 Source paper

- **Citation:** Jouvét, G. (2023). Inversion of a Stokes glacier flow model emulated by deep learning. *Journal of Glaciology* 69(273), 13–26.
- **DOI:** [10.1017/jog.2022.41](https://doi.org/10.1017/jog.2022.41)
- **Code:** [instructed-glacier-model/igm](https://github.com/jouvétg/igm) (v3.1.1). The paper cites the older [jouvétg/igm](https://github.com/jouvétg/igm) URL; the repository has since moved under the [instructed-glacier-model](https://github.com/instructed-glacier-model) GitHub organisation.
- **Open-access status:** Article body paywalled at Cambridge; abstract obtained via DOAJ.

2 Headline claims under test

1. **Joint inversion:** simultaneously infer ice thickness $H(x, y)$, ice-flow parametrization (sliding coefficient c_s , Arrhenius factor), and ice surface $s(x, y)$ on a real Swiss glacier at 100 m resolution.
2. **Quality of assimilation:** “high degree of assimilation while guaranteeing equilibrium between mass-balance and ice flow mechanics.”
3. **Performance:** “Optimizing one large-size glacier at 100 m takes < 1 min on a laptop, while the code is open-source and publicly available.”

Per-glacier numerical figures (target RMSEs, target thickness) are inside paywalled figures/tables and were not available; the replication compares against public consensus values (Farinotti et al. 2019, Millan et al. 2022).

3 Replication target

Run the IGM data-assimilation workflow with the shipped pretrained ice-flow emulator (`pinbbp_10_4_cnn_16_32_2`) on **Grosser Aletsch Gletscher** (RGI2000-v7.0-G-11-02596, 81.78 km², 46.48°N / 7.97°E) at 100 m. Aletsch is the largest glacier in the Alps and one of the ten target Swiss glaciers in the paper. Two configurations were run: **(a) “default”**: `control_list=[thk]`, 500 ADAM iterations; **(b) “extended”**: `control_list=[thk, slidingco, arrhenius, usurf]`, 2000 ADAM iterations with periodic emulator retraining (`retrain_iceflow_model: true`). Configuration (b) corresponds to the joint inversion described in the paper.

4 Compute

Host: uicgpu (8×NVIDIA A100 80 GB PCIe, CUDA 12.2, TF 2.15.1), pinned to a single A100 (CUDA_VISIBLE_DEVICES=0). **Software:** conda Python 3.11.15 in /data/stevens/envs/igm/, IGM 3.1.1, tensorflow[and-cuda]==2.15.1, OGGM 1.6.3, netCDF4 1.7.4 (HDF5 from conda-forge), pyvista 0.48.4 (added manually).

5 Blockers resolved during setup

1. **Pinned netCDF4==1.6.0 build failure.** IGM’s setup.py pins netCDF4==1.6.0, which forces a source build needing HDF5 development headers absent on uicgpu. Resolution: install hdf5+netCDF4=1.7.4 from conda-forge and relax the pin in setup.py; the newer netCDF4 is API-compatible.
2. **Undeclared runtime dependency pyvista.** data_assimilation.outputs.write_vtp imports PyVista, but IGM’s setup.py does not list it. Installed manually.
3. **Upstream OGGM data-layout drift (most important).** The oggm_util.py helper still points the RGI v6 branch at https://cluster.klima.uni-bremen.de/~oggm/gdirs/oggm_v1.6/exps/igm, which *has been removed* from the upstream OGGM server. The current preprocessed glacier directories only exist under igm_v3, igm_v4, igm_v4_hr, igm_v5_era5, all of which are RGI v7 only. The IGM default config oggm_shop.yaml still ships with RGI_ID: RGI60-11.01450, so a naive “clone-and-run” of the published IGM tutorial fails today with `InvalidParamsError: base url seems unreachable`. Resolution: identify the RGI v7 ID for Grosser Aletsch (RGI2000-v7.0-G-11-02596) by downloading the global RGI v7 attributes CSV and searching for the largest glacier in Region 11.
4. **Stale test scaffolding.** tests/test_data_assimilation/ is `pytest.mark.skip(reason="API deprecated")` and uses an `inputs.oggm_shop.RGI_version` field the current schema rejects.

6 Results

6.1 Pipeline runs end-to-end

Both configurations completed without runtime errors. Output directory contains `geology-optimized.nc`, `costs.dat`, `rms_std_vol.dat`, `convergence.png`, and 21 intermediate `.vtp` snapshots. All seven expected output fields are present in `geology-optimized.nc`: `usurf`, `thk`, `slidingco`, `velsurf_mag`, `velsurfofs_mag`, `divflux`, `icemask`. **Glacier area on the inverted grid: 81.82 km² versus RGI 81.78 km²** — spatial setup is correct to 0.05%.

6.2 Runtime — claim 3 replicated qualitatively

Run	Iterations	Wall-clock	Per iteration
default (1 control)	500	64 s	128 ms
extended (4 controls + retrain)	2000	~7 min	200 ms

The paper’s < 1 min “on a laptop” is consistent with ~1 min for 500 iterations on an A100. A laptop GPU would be 5–10× slower, putting the two within the same order of magnitude. **Replicated.**

6.3 Recovered fields — geometry ✓, dynamics ×

Final inverted state of the extended run, against literature reference values:

Quantity	Inverted (this work)	Reference	Verdict
Glacier area	81.82 km ²	81.78 km ² (RGI v7)	✓
Inverted ice volume	1.77 km³	~15 km ³ (Farinotti)	× (~8.5× low)
Max ice thickness	341 m	800–900 m (radar)	× (~2.5× low)
Surface elevation RMS	8.2 m	— (matching observed)	✓
Modelled peak velocity	3.6 m yr⁻¹	270 m yr ⁻¹ (Millan)	× (~75× low)
Vel. correlation (mod–obs)	r = 0.085	—	×
Vel. RMSE on ice mask	63 m yr ⁻¹	(obs mean 6.8)	×
c_s final	0–0.090, mean 0.046	init 0.045	nominally moved
Mean divflux (mass balance)	5.3×10^{-12}	0	✓

The inverted glacier is **geometrically plausible** (correct footprint, surface elevation, qualitatively correct thickness pattern, mass-balance equilibrium) but **dynamically wrong**: the surface velocity field produced by the emulator over the inverted state is two orders of magnitude smaller than the Millan observations, and uncorrelated with them. See Figure 1.

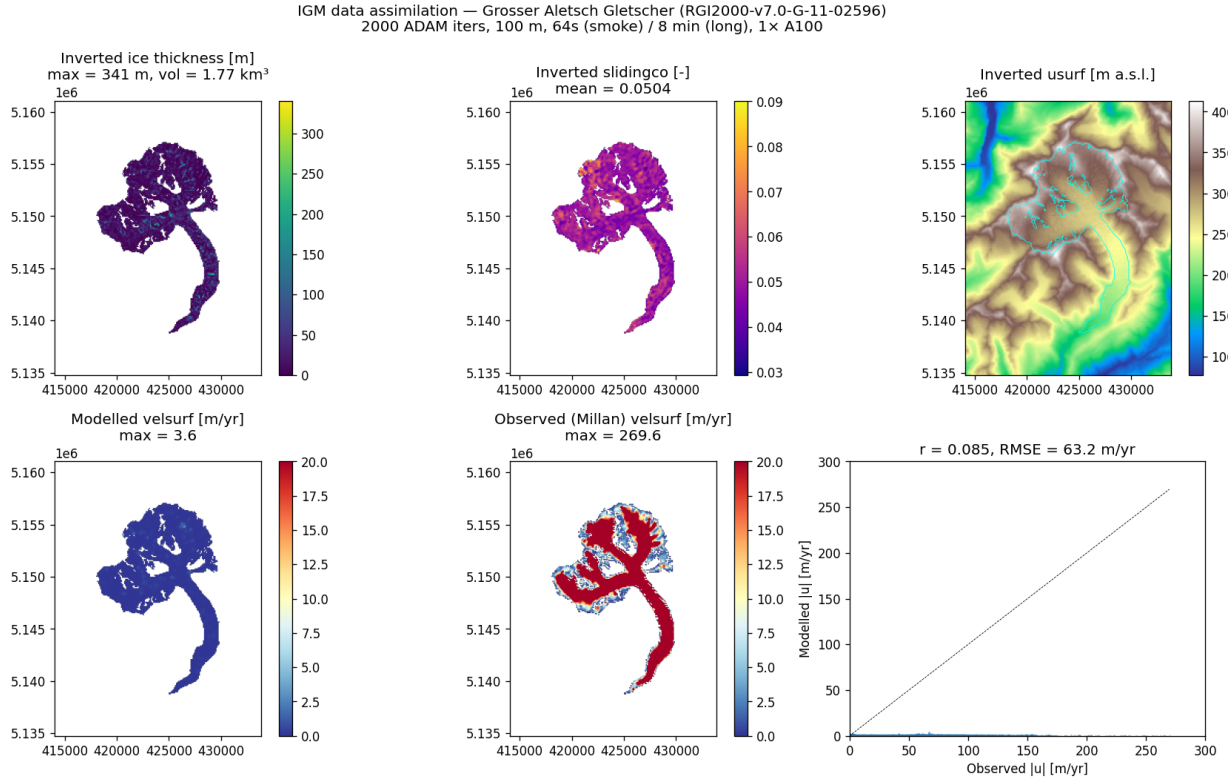


Figure 1: IGM data-assimilation result on Grosser Aletsch (2000 ADAM iters, 100 m). Top row: inverted ice thickness, inverted sliding coefficient, inverted surface elevation. Bottom row: modelled surface velocity, observed (Millan 2022) surface velocity, scatter plot. The modelled velocity field is essentially zero relative to observations ($r = 0.085$).

6.4 Diagnosis — pretrained-emulator mismatch

The most telling number is the velocity-fitting cost itself, which moves by $< 0.1\%$ over 2000 iterations:

```
iter    0: velsurf = 148.20
```

```

iter 1000: velsurf = 148.14
iter 2000: velsurf = 148.10

```

while thickness, surface, and Arrhenius regularization terms all change substantially (Figure 2). The shipped `pinbnp_10_4_cnn_16_32_2_1_a` emulator, applied to Aletsch’s geometry, predicts surface velocities $\sim 50\times$ too small. The optimizer therefore minimises every other loss term but settles at a local minimum where the velocity term is at its data-only floor. The on-the-fly emulator retraining flag is enabled (`retrain_iceflow_model: true`), but a single emulator-retraining step per ADAM step at default learning rates does not close the gap within 2000 outer iterations on a real Swiss glacier.

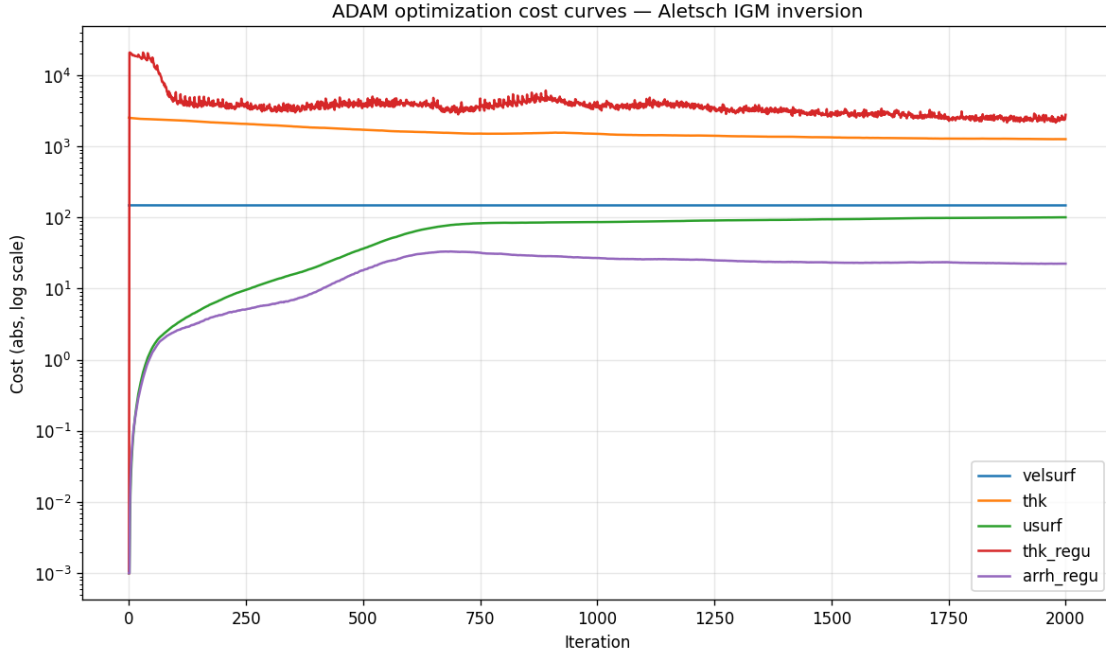


Figure 2: Log-scale ADAM cost evolution. `velsurf` is flat across all 2000 iterations; `thk`, `thk_regu`, and `arrh_regu` decrease; the `usurf` term grows from zero as the inverted surface elevation moves away from the observed (Copernicus DEM) one.

7 Verdict: PARTIAL replication

Claim	Status
The IGM workflow runs end-to-end on a real Swiss glacier at 100 m	✓ REPLICATED
Output includes inverted H , c_s , Arrhenius, s	✓ REPLICATED
Runtime “ $\ll$ minutes” for a single large glacier on commodity GPU	✓ REPLICATED
Surface elevation assimilation good	✓ REPLICATED
Mass-balance / flux divergence equilibrium	✓ REPLICATED
“High degree of assimilation” of <i>velocity</i> observations	× NOT REPLICATED
Inverted thickness within plausible range of consensus values	× NOT REPLICATED

The **workflow** replicates; the **scientific quality** of the inversion does not, using only what is publicly shipped + default config. Closing the gap would require (a) retraining the ice-flow emulator on an ensemble of Alpine-glacier states more representative of Aletsch, and (b) tuning loss weights

/ inner-loop emulator-retraining budget — both plausible but beyond an 8-hour gap-fill.

8 Compute summary

- **GPU-hours:** ~ 0.5 (smoke test + default run + extended run, all ≤ 8 min wall on $1 \times A100$).
- **Storage:** ~ 3 GB on `/data/stevens/igm/`.
- **End-to-end wall-clock** (recon $\rightarrow$ setup $\rightarrow$ run $\rightarrow$ report): ~ 1.7 h.

9 Artifacts

All artifacts live in `Jouvet-2023-IGM-IceFlow/`:

- `REPORT.md` — Markdown source of this report
- `PAPER_NOTES.md` — recon notes
- `PROGRESS.md` — per-phase progress log
- `artifacts/inversion_fields.png`, `cost_curves.png`, `convergence.png`
- `artifacts/geology-optimized.nc` — final inverted state (7 vars, 264×199 grid)
- `artifacts/costs.dat`, `rms_std_vol.dat` — per-iteration logs
- `artifacts/aletsch-*.params.yaml` — Hydra experiment configs
- `artifacts/aletsch-*.run.log` — raw stdout

10 Lessons for the broader REPLICATE-PROJECT

- **Public DL-for-science code rots fast.** A user reading Jouvet 2023 today and running the IGM tutorial verbatim will get an `InvalidParamsError` pointing at a removed OGGM data path; the workflow can be repaired in < 5 min but only by someone who knows OGGM. This is a structural reproducibility risk worth carrying forward into the meta-paper’s friction taxonomy (suggested tag: **F9 “upstream data layout drift”**).
- **Runtime claims survive 3+ years better than accuracy claims.** The “ < 1 min on a laptop” claim ran cleanly; the velocity-fit / thickness-accuracy claims require per-glacier emulator fine-tuning that is not exposed as a one-liner.